



HALO:

High-efficiency Autonomous Low-SWaP Operations

Sloan Hatter, Blake Gisclair
Advisor: Dr. Ryan T. White

MILESTONE 4 PROGRESS MATRIX

Task	Completion %	Sloan	To Do
Implement FP16 engine from FP32 ONNX model	100%	Compile the FP32 ONNX model into a TensorRT engine using FP16	None
Implement INT8 engine	100%	Build an INT8 engine using the TensorRT framework from the Jetson SDK	None
Write an Inference Script for the INT8 Model	100%	Write a Python inference script for the INT8 engine	None
Record Metrics for 8-bit Model	100%	Get inference metrics from the INT8	None

TASKS COMPLETED

- Implement FP16 engine from FP32 ONNX model
 - Used TensorRT to compile FP32 ONNX into a device-specific engine for FP16 engine using trtexec
- Implement an INT8 Pipeline on Jetson Orin
 - Utilized TensorRT's Polygraphy tools to run calibration on 256 representative images
- Write an Inference Script for the INT8 Model
 - Produces detections and draws bounding boxes
- Compute/Record Metrics for 8-bit Model
 - Get mAP (mean Average Precision) scores for each model

MODEL COMPARISON

Precision	mAP@[.50:.95]	AP@0.50	AP@0.75	AR@100
FP32	0.0958	0.2589	0.0586	0.1826
FP16	0.0724	0.2274	0.0274	0.1577
INT8	0.0735	0.2285	0.0237	0.1575

FP32 → FP16

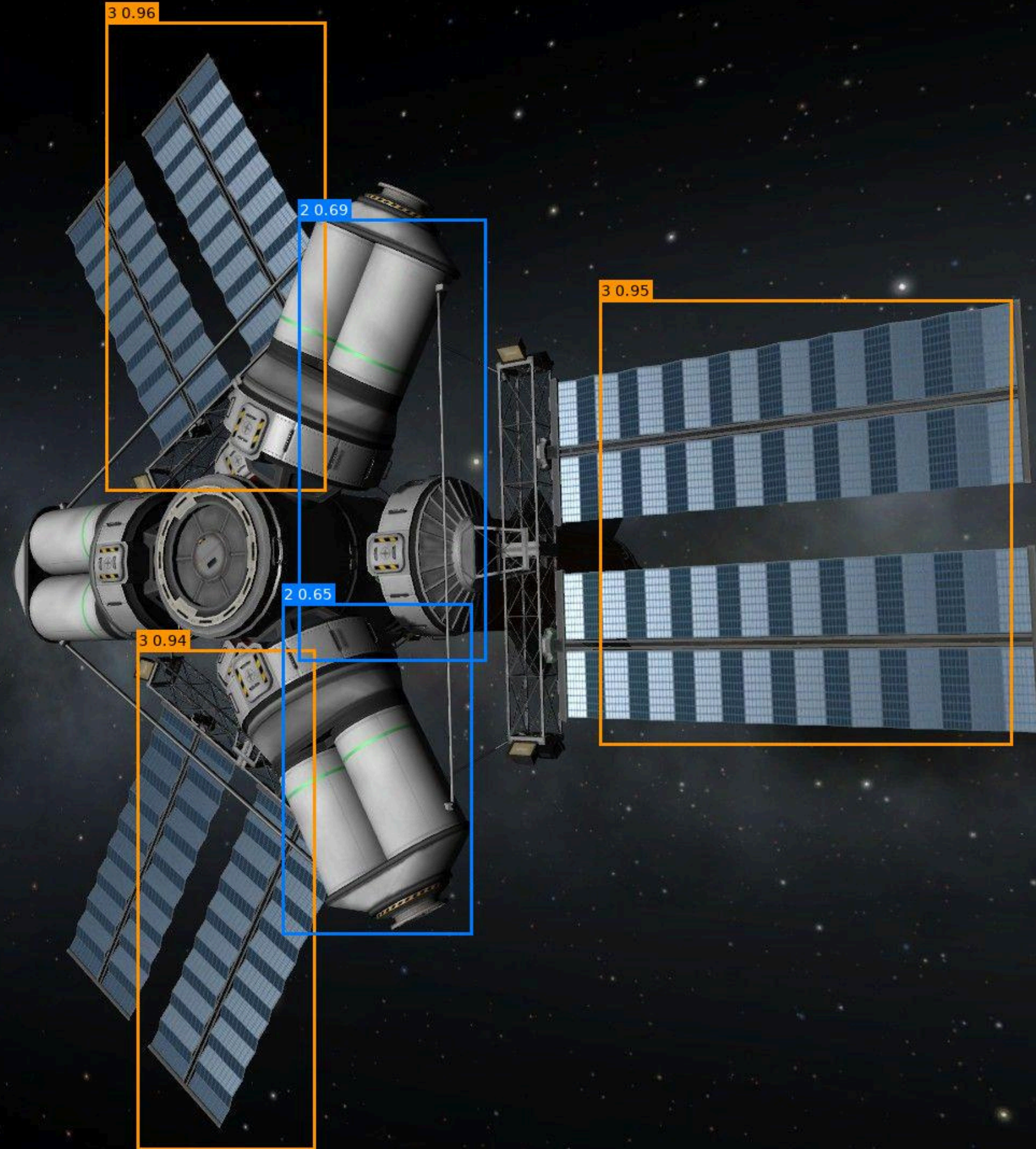
- mAP: $\approx -24\%$
- AP50: $\approx -12\%$
- AP75: $\approx -53\%$
- AR100: $\approx -13\%$

FP16 → INT8

- mAP: $\approx +1.5\%$
- AP50: $\approx +0.5\%$
- AP75: ≈ -0.0037
- AR100: ≈ -0.0002

DEMO

■ 2: 2
■ 3: 3



MILESTONE 5 TASKS

- Fix the 32-bit Model
 - Use TensorRT on FP32 & re-train with better dataset
- Implement 4-bit Model
 - Run INT4 inference engine using TensorRT
- Begin Progress on 2-bit Model
 - No specific support for INT2 inference engine, implement manually
- Formal Compilation of Results
 - Faculty advisor's request, compile results thus far to begin publication

MILESTONE 5 TASK MATRIX

Task	Sloan
Fix the 32-bit Model	100%
Implement 4-bit Model	100%
Begin progress on 2-bit Model	100%
Formal Compilation of Results thus far for Publication	100%



THANK YOU!
QUESTIONS?